DIF2208

Integrales de funciones trigonométricas.

Integrales de Algunas potencias de funciones trigonométricas.

c) Multiplicación de potencias de coseno y seno.

* Distinción entre potencias de grado par v/s de grado impar.

$$\int \cos^h(x)\, \text{sen}^m(x)\, dx.$$

a) h o m es impar, o ambos:

Considere:

$$\text{sen}^2(x) + \cos^2(x) = 1$$

Ejemplo:

$$= \int \cos^5(3x)\, \text{sen}^8(3x)\, dx$$

$$= \int \cos(3x)\cos^4(3x)\,\text{sen}^8(3x)\, dx$$

$$= \int (1-\text{sen}^2(3x))^2\, \text{sen}^8(3x)\cos(3x)\, dx$$

$u = \text{sen}(3x)$, $du = 3\cos(3x)\,dx$

$\frac{dv}{3} = \cos(3x)\,dx$

$$I = \frac{1}{3}\int (1-u^2)^2 u^8\, du = \frac{1}{3}\int (1-2u^2+u^4)u^8$$

ej: $\int \cos^4(x)\,\text{sen}^3(x)\, dx$

$= \int \cos(x)\,\text{sen}^2(x)\,\text{sen}(x)$

$= \int \cos^4(x)(1-\cos^2(x))\,\text{sen}(x)\, dx$

Se resuelve por sustitución simple.

$u = \cos(x)$

$-du = \text{sen}(x)\,dx$

$= -\int u^4(1-u^2)\, du$

$= -\int u^4 - u^6\, du$

$= -\int u^4 du + \int u^6 du =$

Sea ha hay impar (ambos par)

b) Ya y n son pares: Uso identidades.

$$\cos^2(x) = \frac{1}{2} + \frac{\cos(2x)}{2} \quad ; \quad \sin^2(x) = \frac{1}{2} - \frac{\cos(2x)}{2}$$

$$\int \cos^2(x)\,\text{sen}^2(x)\,dx = \int \left(\frac{1}{2} + \frac{\cos(2x)}{2}\right) + \left(\frac{1}{2} - \frac{\cos(2x)}{2}\right) dx$$

$$= \int \left(\frac{1}{4} - \frac{\cos^2(2x)}{4}\right) dx = \int \frac{1}{4}\,dx - \frac{1}{4}\int \cos^2(2x)\,dx$$

$$= \frac{x}{4} - \frac{1}{4}\int \left(\frac{1}{2} + \frac{\cos(4x)}{2}\right) dx$$

$$= \frac{x}{4} - \frac{1}{4}\cdot\frac{x}{2} - \frac{1}{8}\int \cos(4x)\,dx = \frac{x}{4} - \frac{x}{8} - \frac{1}{8}\cdot\frac{\sin(4x)}{4} + C$$

Potencias pares de la tangente

$$\tan^2\alpha = \sec^2\alpha - 1$$

a) $\int \tan^2(x)\,dx$

$$= \int \sec^2(x) - 1\,dx = \int \sec^2(x)\,dx - \int 1\,dx = \tan(x) - x + C$$

b) $\int \tan^4(x)\,dx = \int \tan^2(x)\tan^2(x)\,dx$

$= \int (\sec^2 - 1)(\tan^2(x))\,dx$

$= \int \tan^2(x)\sec^2(x)\,dx - \int \tan^2(x)\,dx$ $\quad u = \tan(x) \rightarrow du = \sec^2(x)\,dx$

$= -(\tan(x) - x) + \int u^2\,du = \frac{u^3}{3} + \tan(x) + x = \frac{\tan^3(x)}{3} + \tan(x) + x$

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$\sec^2 x = \tan^2 x + 1$

Potencias pares de la secante.

1) $\int \sec^2 x\,dx = \tan(x) + C$

2) $\int \sec^4 x\,dx = \int \sec^2 x \cdot \sec^2 x\,dx = \int (\tan^2 x + 1)\sec^2 x\,dx$

$u = \tan x$

$du = \sec^2 x\,dx$

$= \int (u^2 + 1)\,du = \frac{u^3}{3} + u = \frac{\tan^3 x}{3} + \tan x + C$

$\sin^2(x) = \frac{1}{2} - \frac{\cos(2x)}{2}$

3) $\int \sin^2(x)\cos^4(x)\,dx = \int \left(\frac{1}{2} - \frac{\cos(2x)}{2}\right)\left(\frac{1}{2} + \frac{\cos(2x)}{2}\right)^2 dx$ $\cos^2(x) = \frac{1}{2} + \frac{\cos(2x)}{2}$

$= \int \left(\frac{1}{4} - \frac{\cos^2(2x)}{4}\right)\left(\frac{1}{2} + \frac{\cos(2x)}{2}\right) = \int \left(\frac{1}{8} + \frac{\cos(2x)}{8} - \frac{\cos^2(2x)}{8} + -\frac{\cos^3(2x)}{8}\right)dx$

$= \frac{1}{8}\int \left(1 + \cos(2x) - \left(\frac{\cos^2(4x)}{2} + \frac{1}{2}\right) - \cos^2(2x)\cos(2x)\right)dx$

$= \frac{1}{8}\left[x + \frac{\sin(2x)}{2} - \frac{x}{2} - \left(\frac{\sin(4x)}{8}\right) - \int \cos^2(2x)\cos(2x)\,dx\right)$

$= 1 - \sin(2x)\cos(2x)\,dx$

$\Rightarrow u = \sin(2x)$

$du = 2\cos(2x)\,dx$

$\int u^2 \frac{du}{2} = \int \frac{1}{2}\left(1 - u^2\right)du$

$= \int$

$$\int \sec^{14}(x)\,dx = \int \sec^{12}(x)\sec^2(x)\,dx = \int \sec^{12}(x)\cdot\left(\tan^2(x)+1\right)^6 dx$$

$$= \int \underbrace{\left(\tan^2 x + 1\right)^6}_{u=\tan x}\cdot \underbrace{\sec^2(x)}_{du=\sec^2(x)} = \int (u+1)^6\,du = \int u^{12} + 6u^{10}$$

~~Integrais de expressões quadráticas:~~

9. $$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

2. $$\int \frac{1}{\sqrt{a^2-x^2}} = \arcsen\left(\frac{x}{a}\right) + C$$

$2\sqrt{2}$

- $$\int \frac{1}{x^2+2x+2}\,dx = \int \frac{1}{(x+1)^2+1}\,dx = \int \frac{1}{u^2+1}\,du = \arctan(u)+C$$

$u = x+1$
$du = 1\,dx$

$= \arctan(x+1) + C$

- $$\int \frac{1}{x^2-6x+13}\,dx = \int \frac{5}{(x-3)^2+2^2} = \frac{1}{u^2+1} = 5\cdot\frac{1}{2}\arctan\left(\frac{x-3}{2}\right)+C$$

$u = x-3$
$du = 1\,dx$

$x^2-6x+13 =$

$x^2-6x = -13 \;/\; +\left(\frac{6}{2}\right)^2$

$x^2-6x+9 = -4$

$(x-3)^2+4 = 0$

$$\int \frac{1}{\sqrt{6x - x^2 + 5}}\,dx = \int \frac{1}{\sqrt{-(x^2 - 6x + 5)}}\,dx = \int \frac{1}{\sqrt{-[(x-3)^2 - 14)}}\,dx$$

$$= \int \frac{1}{\sqrt{-(x-3)^2 + 14}}\,dx$$

$$x^2 - 6x = +5$$

$$x^2 - 6x + 9 = 14$$

$$(x-3)^2 - 14 = 0$$

$$= \int \frac{1}{\sqrt{-u^2 + 14}}\,du = \arcsen \frac{u}{\sqrt{14}}$$

$$= \arcsen \frac{-x+3}{\sqrt{14}} +$$